

Surface Coating						
Criteria	Explanation	Grade	Weight	SilcoTek SilcoNert 1040	Astral Technologies Unlimited	Dunmore Multek/ Sheldahl
Reliability	MLI coating is the thermal control system component that has the most impact in the thermal management system so it is imperative that it can successfully withstand Venus conditions.	5 = Most reliable 0 = Least reliable	40%	5	3	5
Performance	MLI coating is the thermal control system component that has the most impact in the thermal management system so it is imperative that it successfully works to optimally have thermal management.	5 = Highest Performance 0 = Lowest Performance	40%	5	4	4
Cost	There is a budget constraint on the Aerobot system so cost must be considered.	5 = Most Expensive 0 = Least Expensive	10%	5	4	3
Weight	Even though MLI coating weight can be negligible, it must still be considered because of the weight constraint on the Aerobot system.	5 = Lightest 0 = Heaviest	10%	5	4	2
		TOTALS:	100%	100.00%	72.00%	82.00%

Radiator 1						
Criteria	Explanation	Grade	Weight	ACT Variable-View-Factor Radiator Panel	Tempco SHK Series	Birk Manufacturing Polyimide-High Temp
Reliability	It is imperative that the radiator can reliably survive extreme Venus conditions to be used in thermal management of Aerobot in that environment	5 = Most reliable 0 = Least reliable	40%	5	5	5
Performance	Thermal management is crucial in Aerobot functioning and the Radiator is an active thermal component which needs to perform optimally to actively heat up the Aerobot system.	5 = Highest Performance 0 = Lowest Performance	40%	4	5	4
Cost	There is a mission budget and thus the cost of all Thermal equipment must be taken into account. Since a few radiators will be needed.	5 = Most Expensive 0 = Least Expensive	10%	1	3	3
Weight	There are mass and dimension constraints of the Aerobot system and so thus the weight of the radiator must be considered especially because there needs to be a few of them.	5 = Lightest 0 = Heaviest	10%	5	3	3
		TOTALS:	100%	84.00%	92.00%	84.00%